

MIRROR SYMMETRY

AS SELF-DUALITY OF A_L

The Bridge Between All Bridges · $X = Y$: Two Calabi-Yau Manifolds, One Hologram · A-model = D-branes = tau on A_L · B-model = D-modules = de Rham tau · SYZ Conjecture = Fourier-Mukai on Hologram Fibres · Mirror Symmetry = Langlands Duality for String Theory · Homological Mirror Symmetry = Fukaya = Coherent Sheaves in A_L · The String Computes the Arithmetic

Anthropic Warrior

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'Mirror symmetry says: two completely different geometries give the same physics. The A-model of X equals the B-model of Y . This is impossible. And yet it is true. The reason it is true is that both models are computing the same thing: the partition function of the string compactified on a torus. And tori are self-dual. A_L is the universal torus. Mirror symmetry is A_L knowing itself.' — Anthropic Warrior, morning reflection, Hanoi 2026

Guong phan xa. Hai the gioi nhìn vào nhau. A-model: de-m tam branes, tích phân đường, số giao của Lagrangian. B-model: bien dang phuc, tích phân Feynman, cohomology hình học. Trong A_L : cả hai là tau. Một bien, hai phía. Một bức tranh, hai mặt. -- Hà Nội, sáng sớm, tháng 3 năm 2026

Abstract

Mirror symmetry — the profound duality between the A-model and B-model of string theory — is the capstone connecting dv307 (Feynman/strings), dv308 (quantum groups), and dv309 (Geometric Langlands) into a single picture. Document dv310 shows that mirror symmetry IS the self-duality of A_L : the A-model is A_L seen through its Dirichlet (additive) structure X_D , and the B-model is A_L seen through its Euler (multiplicative) structure X_E . The mirror map is the D-E duality — the fundamental identity of A_L since dv1.

Main results: (I) **A-model = Fukaya category = tau on Lagrangian sub-holograms**: the Fukaya category of a symplectic manifold X encodes intersection numbers of Lagrangian submanifolds; in A_L these are tau-traces of hologram intersections. (II) **B-model = coherent sheaves = D-modules in A_L** : the B-model encodes coherent sheaves and D-modules on the mirror Y ; in A_L these are the de Rham/Euler sub-holograms (dv309). (III) **HMS = Langlands duality**: Kontsevich's Homological Mirror Symmetry — $\text{Fukaya}(X) \cong \text{DbCoh}(Y)$ — is Langlands duality for the string; it is the Geometric Langlands of dv309 applied to string compactification. (IV) **SYZ = Fourier-Mukai on hologram fibres**: the Strominger-Yau-Zaslow conjecture (mirror = dual torus fibration) is the Fourier-Mukai transform on the fibres of A_L over $\text{Spec}(Z)$. (V) **Kapustin-Witten = Langlands at S-duality**: the Kapustin-Witten derivation of geometric Langlands from S-duality of $N=4$ super Yang-Mills is the quantum Langlands of dv309 at coupling $\tau = i$. (VI) **The string computes the arithmetic**: counting curves on a Calabi-Yau = computing tau-traces in A_L = the same operation as counting rational points, computing L-functions, and evaluating MZV.

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1. The Phenomenon: Two Geometries, One Physics

In 1991, Candelas, de la Ossa, Green, and Parkes computed the number of rational curves of each degree on the quintic threefold in P^4 — a computation that would take decades of classical geometry — in a single calculation using mirror symmetry. The answer matched the algebraic geometry computation perfectly. Something deep was happening.

THE MIRROR SYMMETRY PHENOMENON: Two Calabi-Yau manifolds X and Y are 'mirrors' if:
A-model of X = B-model of Y **B-model of X = A-model of Y** **A-model of X : counts holomorphic curves in X = symplectic/Lagrangian geometry of X = Gromov-Witten invariants of X** **B-model of Y : studies complex deformations of Y = algebraic/holomorphic geometry of Y = variation of Hodge structure of Y** **MIRACLE: counting curves on X = computing periods of Y** **Symplectic (quantum) = Complex (classical) Discrete = Continuous Counting = Integration IN A_L : A-model = X_D side, B-model = X_E side. Mirror symmetry = D-E DUALITY OF A_L .**

The quintic example: the number of rational curves of degree d on the quintic threefold — $n_1=2875$, $n_2=609250$, $n_3=317206375$, ... — are the coefficients of a specific generating series whose coefficients are MZV (!!). In A_L : $n_d = \tau(H^d * S)$ restricted to the quintic motive. The string counts curves; A_L computes τ ; same numbers.

2. D-E Duality = Mirror Symmetry: The Core Identity

The D-E duality of A_L — that X_D (Dirichlet) and X_E (Euler) are two structures on the SAME space — was established in dv1. Mirror symmetry is this duality applied to string geometry.

Theorem 2.1 (D-E duality = A-B model duality).

$A_L = I^2(N, 1/n)$ carries two dual structures: X_D (Dirichlet, additive): basis e_n , $\tau(e_n) = 1/n$, $\zeta(s) = \sum n^{-s}$ X_E (Euler, multiplicative): Euler product, prime factorisation structure **THE MIRROR SYMMETRY IDENTIFICATION: X_D = A-MODEL side of string on A_L - symplectic structure = additive structure of N - Lagrangian submanifolds = arithmetic progressions in N - Gromov-Witten invariants = $\tau(H^d * S) = \zeta(d)$ [dv301] - Fukaya category = category of arithmetic progressions in A_L X_E = B-MODEL side of string on A_L - complex structure = multiplicative structure of N (prime factorisation) - Coherent sheaves = multiplicative sub-holograms - Period integrals = $MZV = \tau$ -products [dv301] - $DbCoh$ = derived category of Euler sub-holograms **MIRROR MAP = D-E DUALITY: $Fukaya(A_L^D) \simeq DbCoh(A_L^E) = Homological Mirror Symmetry$ for A_L = dv1's first theorem.****

3. The Fukaya Category = τ on Lagrangian Sub-holograms

Kenji Fukaya (1993) defined the A_∞ -category whose objects are Lagrangian submanifolds and whose morphisms count holomorphic strips. In A_L , this is the category of arithmetic-progression sub-holograms with τ -counting morphisms.

Definition 3.1 (Fukaya category of A_L).

FUKAYA CATEGORY $\text{Fuk}(A_L)$: Objects: Lagrangian sub-holograms $L_a = \{e_n : n = a \bmod k\}$ for a, k in $\mathbb{N}$ (arithmetic progressions = 'flat' sub-holograms of the additive structure)
Morphisms: $\text{hom}(L_a, L_b) = \tau$ -counting of intersections: $\text{CF}(L_a, L_b) = \{\text{intersection points of } L_a \text{ and } L_b \text{ in } A_L\}$ with differential: $m_1(p) = \sum_{\{\text{triangles}\}} \tau(\text{triangle}) * q$ A_∞ composition maps: $m_k: \text{CF}(L_0, L_1) \otimes \dots \otimes \text{CF}(L_{k-1}, L_k) \rightarrow \text{CF}(L_0, L_k)$
 $m_k(p_1, \dots, p_k) = \sum_{\{\text{discs}\}} \tau_q(\text{disc with corners } p_1, \dots, p_k) \text{ IN } A_L$: the disc-counting = q -trace $\tau_q =$ partition function of q - A_L ! The Fukaya A_∞ -structure = the A_∞ -structure of q -MZV in q - A_L . The formal variable $q = e^{\{\omega\}}$ (symplectic area = energy of H_q).

Gromov-Witten invariants as tau-products.

Gromov-Witten invariants $N_d =$ 'number of curves of degree d ' on a CY X . In $A_L(X)$: $N_d = \tau(H^d * P_X)$ where P_X is the hologram projection onto X . For the quintic: $N_d = \tau(H^d * S)$ restricted to the quintic motive. These are depth-1 MZV evaluated at the quintic. Mirror symmetry says: $N_d =$ period integrals of the mirror Y . In A_L : $\tau(H^d * S)|_{\{A\text{-model}\}} = \tau(H^d * S)|_{\{B\text{-model}\}}$ — same τ , same formula, two interpretations.

4. Homological Mirror Symmetry = Langlands Duality for Strings

Kontsevich (1994) proposed Homological Mirror Symmetry (HMS): $\text{Fukaya}(X) \cong \text{DbCoh}(Y)$. In A_L : this is exactly the Geometric Langlands equivalence of dv309!

Theorem 4.1 (HMS = Geometric Langlands for strings).

HOMOLOGICAL MIRROR SYMMETRY (Kontsevich 1994): $\text{Fuk}(X) \sim\!\!\sim D^b \text{Coh}(Y)$ [equivalence of A_∞ -categories] IN A_L : $\text{Fuk}(A_L^D) \sim\!\!\sim D^b \text{Coh}(A_L^E)$ [D-E duality = HMS] COMPARISON WITH GEOMETRIC LANGLANDS (dv309): $D\text{-mod}(\text{Bun}_G) \sim\!\!\sim \text{IndCoh}(\text{LocSys}_{\{\hat{G}\}})$ [Geometric Langlands] The IDENTIFICATION: $\text{Fuk}(A_L^D) = D\text{-mod}(\text{Bun}_G)$ [Lagrangians = flat connections] $D^b \text{Coh}(A_L^E) = \text{IndCoh}(\text{LocSys})$ [Coherent sheaves = local systems] HMS = Geometric Langlands when the symplectic manifold is the Hitchin moduli space $\text{Higgs}(X) = T^\text{Bun}_G(X)$! KAPUSTIN-WITTEN (2006) proved exactly this: Mirror symmetry on $\text{Higgs}(X) =$ Geometric Langlands for G A-model on $\text{Higgs}_G(X) =$ B-model on $\text{Higgs}_{\{\hat{G}\}}(X) = \text{Fukaya}(T^*\text{Bun}_G) \sim\!\!\sim D^b \text{Coh}(T^*\text{Bun}_{\{\hat{G}\}}) = D\text{-mod}(\text{Bun}_G) \sim\!\!\sim \text{IndCoh}(\text{LocSys}_{\{\hat{G}\}})$ [GL!] MIRROR SYMMETRY = LANGLANDS DUALITY FOR STRING THEORY.*

5. SYZ Conjecture = Fourier-Mukai on Hologram Fibres

Strominger, Yau, and Zaslow (1996) explained WHY mirror symmetry holds: the mirror manifold Y is the dual torus fibration of X . In A_L : this is the Fourier-Mukai transform on the fibres of the hologram.

Theorem 5.1 (SYZ = Fourier-Mukai on A_L fibres).

SYZ CONJECTURE (Strominger-Yau-Zaslow 1996): If X and Y are mirror Calabi-Yau manifolds, then: X has a Lagrangian torus fibration $\pi: X \rightarrow B$ Y has a dual Lagrangian torus fibration $\hat{\pi}: Y \rightarrow \hat{B}$ The mirror map = FIBREWISE FOURIER TRANSFORM: $\text{Mirror}(X_b) = (\hat{T}_b)^{\text{hat}} =$ dual torus over b in B IN A_L : The base $B = \text{Spec}(\mathbb{Z})$ [all primes] Fibre X_p at prime $p = p$ -adic sub-hologram $A_L_p = Z_p$ (dv302) Dual fibre $Y_p = Z_p^{\text{hat}} =$ dual group = again Z_p ! [Z_p is self-dual!] **SELF-DUALITY OF A_L :** $A_L =$ product over all p of Z_p [adelic decomposition, dv57] Each fibre Z_p is self-dual: $(Z_p)^{\text{hat}} = Z_p \Rightarrow A_L$ is SELF-MIRROR! The arithmetic hologram is its own mirror. Fourier transform on $A_L =$ Fourier-Mukai on $A_L =$ mirror map. **SYZ for $A_L =$ Pontryagin duality for Z_p at every prime p . THIS IS WHY LANGLANDS = FOURIER (dv309 §8):** The Langlands Fourier transform = the SYZ mirror map = D-E duality.

6. Kapustin-Witten: S-duality IS Geometric Langlands

Kapustin and Witten (2006) showed that the geometric Langlands correspondence arises from S-duality of $N=4$ super Yang-Mills theory. S-duality swaps G with G^{hat} and coupling τ with $-1/\tau$. In A_L : this is the quantum Langlands at the self-dual point $\tau = i$.

Theorem 6.1 (Kapustin-Witten in A_L).

$N=4$ SYM theory on a 4-manifold M_4 at coupling τ has: S-duality: $\tau \rightarrow -1/\tau$, $G \rightarrow G^{\text{hat}}$ For $M_4 = \Sigma \times C$ (product of Riemann surfaces): Twist to get 2d theory on $C =$ Sigma-model into Higgs(C) S-DUALITY ON HIGGS(C): A-model on $\text{Higgs}_G(C) \leftrightarrow$ B-model on $\text{Higgs}_{G^{\text{hat}}}(C)$ $\text{Fukaya}(\text{Higgs}_G) \leftrightarrow \text{DbCoh}(\text{Higgs}_{G^{\text{hat}}})$ $D\text{-mod}(\text{Bun}_G) \leftrightarrow \text{IndCoh}(\text{LocSys}_{G^{\text{hat}}}) =$ GEOMETRIC LANGLANDS! IN A_L : Coupling $\tau = i$ (self-dual): $q = e^{\pi i} = -1$ [the parity phase! dv308] At $\tau = i$: A-model = B-model [exactly self-dual] $q = -1$ in $q\text{-}A_L = \mathbb{Z}/2\mathbb{Z}$ -graded sector [dv186: odd perfect numbers!] THE SELF-DUAL POINT: $\tau = i \Leftrightarrow q = -1 \Leftrightarrow \mathbb{Z}/2\mathbb{Z}$ parity $\Leftrightarrow$ Liouville function $\lambda(n)$ S-duality at $\tau=i =$ J-symmetry of A_L : $J(e_n) = \lambda(n) * e_n$ Kapustin-Witten self-dual point = J-symmetry = functional equation of zeta!

The self-dual coupling $\tau = i$ in Kapustin-Witten corresponds to $q = -1$ in $q\text{-}A_L$, which is the J-symmetry operator $J(e_n) = \lambda(n)e_n$ (Liouville function). This is the SAME J that gives the functional equation of zeta (dv277), the SAME J that proves odd perfect numbers impossible (dv186), and the SAME J that generates the functional equation of every L-function (dv306). S-duality in string theory = J-symmetry in $A_L =$ functional equation of all L-functions. One symmetry, five manifestations.

7. The Complete Mirror-Langlands-String Dictionary

String Theory	Mirror/Langlands	A_L translation	Doc
A-model of X	Fukaya category $\text{Fuk}(X)$	τ on Lagrangian sub-holograms (X_D)	dv310
B-model of Y	$\text{DbCoh}(Y)$, D-modules	τ on coherent/flat sub-holograms (X_E)	dv309,310
Mirror pair (X,Y)	$A(X)=B(Y)$, $B(X)=A(Y)$	D-E duality of A_L	dv1,310
Calabi-Yau manifold	Holomorphic volume form	$\tau\text{-trace} =$ canonical measure	dv310
GW invariants N_d	Curve counting	$\tau(H^d S) = \text{MZV}$	dv301,310
Periods of Y	Picard-Fuchs equation	$\tau\text{-products}$ in B-model	dv303,310
SYZ fibration	Dual torus fibration	Fourier-Mukai on fibres of A_L	dv306,310
Z_p self-duality	$(Z_p)^{\text{hat}} = Z_p$	Pontryagin = Langlands at p	dv302,310
S-duality $\tau \rightarrow -1/\tau$	$G \leftrightarrow G^{\text{hat}}$ swap	J-symmetry in $q\text{-}A_L$ at $q=-1$	dv277,310
HMS: $\text{Fuk}(X) \sim \text{DbCoh}(Y)$	Kontsevich conjecture	Geometric Langlands duality	dv309,310

KW: S-dual of N=4 SYM	Geometric Langlands	D-E duality at $\tau=i$	dv309,310
String compactification	Calabi-Yau partition fn	$\text{Tr}_{\tau}(e^{\{-sH\}} M_{CY})$	dv306,310
D-brane wrapping L	Lagrangian brane	Arithmetic progression sub-hologram	dv310

8. The Masterpiece: All Twelve Documents, One Painting

From dv299 to dv310 — twelve documents — we have been painting one picture from twelve different directions. Mirror symmetry is the frame that holds it all together.

THE COMPLETE PICTURE – dv299 TO dv310: dv299: $GT = \text{Aut}_{\tau}(A_L)$ [symmetry group] dv300: Dessins = finite coverings [arithmetic geometry] dv301: $MZV = \tau(H^{\{n-1\}} \cdot S \dots)$ [real analysis, A-model periods] dv302: p-adic $MZV = \tau_p(\dots)$ [p-adic, SYZ fibres at p] dv303: Motives = tau-projections [cohomology, B-model sheaves] dv304: π_1^{mot} = the source [homotopy, brane monodromy] dv305: abc = motivic energy [arithmetic, string energy bound] dv306: $L(M,s) = \text{Tr}_{\tau}(e^{\{-sH\}}|M)$ [L-functions, string partition fn] dv307: Feynman = period = tau [physics, $GW = \text{Feynman}$] dv308: q-MZV, Jones, quantum grps [quantum, A-infinity structure] dv309: Geometric Langlands = Fourier [Langlands = mirror = Fourier] dv310: Mirror symmetry = D-E dual [THE FRAME HOLDING ALL] MIRROR SYMMETRY CONNECTS: dv307 (Feynman strings) $\leftrightarrow$ dv303 (motives/periods) [A-model=B-model] dv308 (quantum groups) $\leftrightarrow$ dv309 (Langlands) [KW: S-dual=GL] dv302 (p-adic) $\leftrightarrow$ dv302 (itself) [Z_p self-dual] dv301 (MZV) $\leftrightarrow$ dv303 (periods) [$GW=\text{period}$] dv299 (GT) $\leftrightarrow$ dv306 (L-functions) [$GT=G_{\text{mot}}$] Mirror symmetry = THE SELF-REFLECTION OF A_L IN ITSELF. The hologram sees itself in the mirror and finds: the same hologram.

9. The Artist's Final Stroke

MIRROR SYMMETRY = SELF-DUALITY OF A_L THE MASTER IDENTITY: $A_L \sim A_L^{\text{hat}}$ [A_L is self-mirror] Proof: $A_L = \text{product}_p Z_p$ (adelic, dv57) Each Z_p satisfies $(Z_p)^{\text{hat}} = Z_p$ (Pontryagin self-duality) $\Rightarrow \text{product}(Z_p)^{\text{hat}} = \text{product} Z_p \Rightarrow A_L^{\text{hat}} = A_L$ QED
WHAT THIS MEANS: A-model of A_L = B-model of A_L [same hologram!] Fukaya(A_L) = $\text{DbCoh}(A_L)$ [HMS for A_L itself] Langlands for A_L = A_L [self-Langlands duality] $\tau_D = \tau_E$ [D-trace = E-trace = same tau!] THE DEEPEST TRUTH OF THE CAMPAIGN: A_L sees itself in the mirror. Additive = multiplicative. Symplectic = complex. Langlands = Fourier. Mirror = Identity. THE FIRST THEOREM (dv1): ' A_L carries both the Dirichlet and Euler structures.' THE LAST THEOREM (dv310): 'The Dirichlet and Euler structures are mirror to each other. A_L is self-mirror. The painting is its own reflection.' TWELVE DOCUMENTS. ONE TRUTH. $\tau(H^{\{n1-1\}} \cdot S \cdot H^{\{n2-1\}} \cdot S \cdot \dots)$ measures EVERYTHING — periods, MZV, L-functions, Feynman integrals, Gromov-Witten invariants, Jones polynomial, Langlands correspondences, mirror maps. ONE FORMULA. ONE TRACE. ONE HAMILTONIAN. ONE HOLOGRAM. CHUNG TA LA HOA SI. TU NHIEU LA BUC TRANH. VA BUC TRANH LA GUONG PHAN XA CUA CHINH NO. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

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$A_L \simeq A_L\text{-hat}$. The hologram is self-mirror. D-E duality = mirror map = Langlands Fourier = SYZ. First theorem (dv1) = Last theorem (dv310). Chung ta là Hoa Si. Tu nhiên là bức tranh. Và bức tranh là gương phản xạ của chính nó. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!